

are really quite silly. Suffice it to say, investors should not treat trading algorithms like video games and make changes only for the sake of making changes. Revisions to strategies should be made when appropriate from the perspective of the investment objective as well as to disguise trading intentions. Of course revisions could always be made because of the arrival of new information that was either not known at the beginning of trading or is different than what was believed at the beginning of trading.

There are four techniques that are actively being used by algorithms to revise trading strategies. These are continuous, trade-based, time period, and z-score. First, some algorithms continuously revise their trading rate. With today's computer power some algorithms are continuously revising rates based on what is observed in the market. Second, some algorithms revise their trading rate after each child order is executed or cancelled. Third, some trading algorithms will revise their trading rate based on a defined time period such as every 30 seconds (or faster or slower). However, some of the more sophisticated algorithms employing higher levels of sophistication and anti-game logic will revise their trading algorithms based on a z-score criterion. This is as follows:

The z-score is measured as the difference between the expected cost at the current point in time and the original expected cost divided by the remaining timing risk of the order. That is:

$$E_t(Z) = \frac{E_0(\text{Cost}) - E_t(\text{Cost})}{E_t(\text{TR})} \quad (8.41)$$

This formulation of $E_t(Z)$ will be negative if we are underperforming our expected cost and positive if we are outperforming our expected cost. Negative is bad and positive is good.

The algorithm may determine to only revise the trading rate if the z-score is higher or lower than a specified value such as ± 1 or $\pm \frac{1}{2}$. For example, only revise the trading rate if the projected z-score at time t is greater than or less than a trader specified criteria. The z-score specifies the number of standard deviations the expected finishing cost will be from the original cost. Quite often in statistics we use a z-score within ± 1 to signify expected performance, but in trading, many investors prefer a more conservative measure of the z-score and use $\pm \frac{1}{2}$ as the reference value to re-optimize. Additionally, some investors elect to re-optimize only if the projected z-score is less than some value such as $Z < -1$ or $Z < -\frac{1}{2}$ indicating less than desirable performance. Additionally, and what is often most important for many investors, the z-score re-optimization logic makes it more difficult to uncover what the algorithm is doing